

SQL CASE STUDY - EMPLOYEE ANALYTICS

Create master tables

Create table Departments(dept_id int primary key,dept_name varchar(50) not null unique);

create table Locations(location_id int primary key,location_name varchar(50) not null unique);

create table Job_Roles(role_id int primary key,role_name varchar(50) not null,min_salary decimal(10,2),max_salary decimal(10,2));

Create Employees Table

create table Employees(emp_id int primary key,emp_name varchar(50) not null,gender varchar(10),
dept_id int,role_id int,location_id int,join_date date,
status varchar(20) default 'Active',
foreign key (dept_id) references Departments(dept_id),
foreign key (role_id) references Job_Roles(role_id),
foreign key (location_id) references Locations(location_id));

Create salaries Table

create table Salaries(salary_id int primary key,
emp_id int,
salary decimal(10,2),
effective_date date
foreign key (emp_id) references Employees(emp_id))

Create performance reviews Table

create table Performance_Reviews(review_id int primary key,
emp_id int,
review_year int,
rating decimal(2,1) check(rating between 1 and 5),
foreign key (emp_id) references Employees(emp_id));

Importing Data from csv files

```
BULK INSERT Departments
FROM 'C:\Temp\departments.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

```
BULK INSERT Locations
FROM 'C:\Temp\locations.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

```
BULK INSERT Job_Roles
FROM 'C:\Temp\jobroles.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

```
BULK INSERT Employees
FROM 'C:\Temp\employees.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

```
BULK INSERT Salaries
FROM 'C:\Temp\salaries.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

```
BULK INSERT Performance_Reviews
FROM 'C:\Temp\performance.csv'
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
);
```

Performance Analysis

1. Top 10 high performing employees

```
select top 10 e.emp_name,p.rating from Employees e join  
Performance_Reviews p  
on e.emp_id = p.emp_id order by p.rating desc;
```

	emp_name	rating
1	Rohan Das	4.9
2	Shilpa B	4.9
3	Suraj K	4.9
4	Amrutha K	4.9
5	Sachith M	4.9
6	Ashwini P	4.9
7	Sudeep R	4.9
8	Teena L	4.9
9	Linda S	4.8
10	Rajiv K	4.8

2.Average rating per department

```
select d.dept_name,avg(p.rating) as Average_Rating from  
Performance_Reviews p join Employees e on p.emp_id =  
e.emp_id  
join Departments d on e.dept_id = d.dept_id group by  
(d.dept_name);
```

	dept_name	Average_Rating
1	Administration	4.280000
2	Business Analytics	4.320000
3	Customer Support	3.960000
4	Development	4.700000
5	Finance	4.440000
6	HR	4.140000
7	IT	4.166666
8	Legal	4.660000
9	Logistics	4.400000
10	Maintenance	4.775000
11	Marketing	4.480000
12	Operations	3.780000
13	Procurement	3.760000
14	Product Manage...	4.060000
15	Public Relations	4.320000

3.Employee eligible for promotion(rating > 4 and 2+ years of experience)

```
select e.emp_id,e.emp_name,p.review_year - year(e.join_date) as  
experience from Employees e join Performance_Reviews p on
```

e.emp_id= p.emp_id where p.rating > 4 and p.review_year - year(e.join_date) > 2;

	emp_id	emp_name	experience
1	102	Priya Nair	4
2	103	Rahul Verma	5
3	105	Arjun Menon	6
4	107	Vikram Singh	7
5	108	Meera Pillai	8
6	111	Rohan Das	9
7	112	Pooja Reddy	5
8	114	Divya Nam...	6
9	116	Lakshmi Nair	7
10	117	Sachin Patil	10
11	118	Aparna Roy	11
12	121	Abhishek J...	12
13	123	Sandeep ...	5
14	125	Gokul Raj	6
15	127	Naveen K	7

4.Performance Trend over year

**select avg(rating) as avg_rating,review_year from Performance_Reviews
group by(review_year);**

	avg_rating	review_year
1	4.226666	2023
2	4.320000	2024
3	4.433333	2025

Salary Analysis

5.Salary Distribution per Departments

**select d.dept_name,sum(s.salary) as total_salary from Departments d
join Employees e on e.dept_id = d.dept_id join Salaries s on e.emp_id =
s.emp_id group by(d.dept_name);**

	dept_name	total_salary
1	Administration	7300000.00
2	Business An...	5700000.00
3	Customer S...	3290000.00
4	Development	14800000...
5	Finance	4690000.00
6	HR	3540000.00
7	IT	4540000.00
8	Legal	13400000...
9	Logistics	7000000.00
10	Maintenance	24900000...
11	Marketing	7400000.00
12	Operations	2480000.00
13	Procurement	3100000.00
14	Product Ma...	4700000.00
15	Public Relat...	7500000.00

6.Highest and lowest salary per role

```
select r.role_name,max(s.salary) as highest_salary,min(s.salary) as  
lowest_salary from Job_Roles r join Employees e  
on r.role_id = e.role_id join Salaries s on e.emp_id = s.emp_id group  
by(r.role_name);
```

	role_name	highest_salary	lowest_salary
1	Analyst	600000.00	400000.00
2	Assistant...	740000.00	650000.00
3	Associat...	2800000.00	2500000.00
4	Director	3500000.00	3000000.00
5	Finance ...	680000.00	550000.00
6	Finance ...	1500000.00	1300000.00
7	HR Exe...	530000.00	450000.00
8	HR Man...	1250000.00	1000000.00
9	Lead	1650000.00	1400000.00
10	Manager	1200000.00	900000.00
11	Marketin...	690000.00	600000.00
12	Marketin...	1600000.00	1400000.00
13	Operatio...	1850000.00	1600000.00
14	Project ...	1850000.00	1700000.00
15	Senior A...	890000.00	750000.00

7. Salary vs Performance Correlation

```
select
  ( count(*) * sum(s.salary * p.rating) - sum(s.salary)*sum(p.rating))/
  sqrt(( (count(*)*sum(s.salary*s.salary) - power(sum(s.salary),2) ) *
    (count(*)*sum(p.rating*p.rating) - power(sum(p.rating),2))
    )) as salary_performance_correlation from Salaries s join
Performance_Reviews p on
s.emp_id = p.emp_id;
```

	salary_performance_correlation
1	0.720774084770709

8.) Employees below department average salary

```
select e.emp_name,e.dept_id,s.salary from Employees e join Salaries s
on e.emp_id = s.emp_id
where s.salary < (
select avg(s1.salary) from Salaries s1 join Employees e1 on s1.emp_id =
e1.emp_id where e.dept_id = e1.dept_id);
```

	emp_name	dept_id	salary
1	Amit Sharma	1	500000.00
2	Priya Nair	1	750000.00
3	Sandeep ...	2	700000.00
4	Sneha Iyer	2	650000.00
5	Ritika Shar...	3	850000.00
6	Sharon E	3	870000.00
7	Sruthi P	3	880000.00
8	Nayana R	3	890000.00
9	Arjun Menon	4	1100000...
10	Neha Gupta	5	450000.00
11	Swathi P	5	480000.00
12	Hemant Y	6	1750000...
13	Naveen K	6	1700000...
14	Anu Thomas	7	2100000...
15	Lekha S	7	2150000...

Department Growth

9. Hiring Trend per year

```
select count(e.emp_id) as employees_hirries,year(e.join_date) as year
from Employees e group by year(e.join_date);
```

	employees_hirres	year
1	3	2010
2	5	2011
3	5	2012
4	5	2013
5	5	2014
6	5	2015
7	10	2016
8	10	2017
9	10	2018
10	10	2019
11	11	2020
12	10	2021
13	10	2022
14	1	2023

10.Headcount Growth by departments

```
select d.dept_name,count(e.emp_id) as  
employees_hiried,year(e.join_date) as year from Departments d  
join Employees e on d.dept_id = e.dept_id  
group by year(e.join_date),d.dept_name;
```

	dept_name	employees_hiried	year
1	Administration	1	2014
2	Administration	4	2019
3	Business An...	5	2016
4	Customer S...	1	2019
5	Customer S...	4	2020
6	Development	4	2014
7	Development	1	2023
8	Finance	1	2018
9	Finance	4	2021
10	HR	4	2018
11	HR	1	2020
12	IT	5	2019
13	IT	1	2021
14	Legal	5	2018
15	Logistics	5	2017

Strategic Insights

11.employees overdue for promotion

```
select e.emp_name,(p.review_year - year(e.join_date)) as  
promotion_overdue from Employees e join Performance_Reviews p  
on e.emp_id = p.emp_id where p.rating > 4;
```

	emp_name	promotion_overdue
1	Amit Sharma	2
2	Priya Nair	4
3	Rahul Verma	5
4	Arjun Menon	6
5	Vikram Sin...	7
6	Meera Pillai	8
7	Rohan Das	9
8	Pooja Reddy	5
9	Divya Nam...	6
10	Lakshmi Nair	7
11	Sachin Patil	10
12	Aparna Roy	11
13	Abhishek J...	12
14	Sandeep ...	5
15	Ritika Shar...	2

12.High salary but low performance cases

```
select e.emp_name,s.salary,p.rating from Salaries s join Employees e on  
s.emp_id = e.emp_id join Performance_Reviews p  
on e.emp_id = p.emp_id where s.salary > (select avg(salary) from  
Salaries)  
and p.rating < (select avg(rating) from Performance_Reviews);
```

	emp_name	salary	rating
1	Pooja Reddy	2800000.00	4.3
2	Hemant Y	1750000.00	4.3
3	Pranav R	1800000.00	4.3

13.location wise salary comparison

```
select l.location_name,avg(s.salary) as average_salary from Locations l  
join Employees e  
on l.location_id = e.location_id join Salaries s on e.emp_id = s.emp_id  
group by(l.location_name) order by avg(s.salary) desc;
```


	location_name	average_salary
1	Bhubaneswar	4640000.000000
2	Bangalore	4120000.000000
3	Jaipur	2800000.000000
4	Nagpur	2620000.000000
5	Kochi	2230000.000000
6	Ahmedabad	2020000.000000
7	Kolkata	1860000.000000
8	Lucknow	1790000.000000
9	Vishakhapat...	1540000.000000
10	Gurgaon	1420000.000000
11	Mumbai	1350000.000000
12	Coimbatore	1220000.000000
13	Noida	1138000.000000
14	Chandigarh	1130000.000000
15	Chennai	918000.000000

14.gender diversity ratio

**select e.gender,count(e.emp_id) as total_number from Employees e
group by e.gender;**

	gender	total_number
1	Female	49
2	Male	51

15.median salary

**select distinct PERCENTILE_CONT(0.5)
within group (order by salary) over() as median_salary from Salaries;**

	median_salary
1	1225000

16.identify leadership pipeline candidate

**select e.emp_name as leadership_candidates from Employees e join
Salaries s on e.emp_id = s.emp_id join Performance_Reviews p on
e.emp_id = p.emp_id
where p.rating >4.5 and s.salary > (select avg(salary) from Salaries);**

	leadership_candidates
1	Vikram Singh
2	Meera Pillai
3	Rohan Das
4	Aparna Roy
5	Abhishek Jain
6	Anu Thomas
7	Vishal Mehta
8	Keerthana R
9	Shilpa B
10	Tarun G
11	Lekha S
12	Suraj K
13	Preethi N
14	Amrutha K
15	Divakar R